2. PROCESS DESCRIPTION

2.1 INTRODUCTION OF SOURCE NATURAL GAS TO THE PLANT

Source natural gas is introduced to the plant at a pressure of 20 Kg/cm^2G to 39 Kg/cm^2G at the battery limit.

After reducing the pressure to about 19 Kg/cm^2G by PICA-001, natural gas as a process feed stock is directly delivered to Natural Gas Compressor (K-303) and as a fuel the pressure is further reduced to about 3 Kg/cm^2G by PICA-101, delivered to the High Pressure Boiler Unit, Primary Reformer (R-101), Fired Gas Heater (H-101) and Start-Up Heater (H-401).

2.2 SYNTHESIS GAS PREPARATION

Synthesis gas preparation is the first stage in the production of ammonia from natural gas, air and steam. The function is the production of the gas suitable for ammonia production, that is, with a hydrogen to nitrogen ratio of three to one, with a minimum of residual methane.

a. Desulfurization

The raw natural gas feed stream contains sulfur compounds in the form of hydrogen sulfide and mercaptane.

These forms of sulfur compounds are poisonous to the catalyst in the Primary and Secondary Reformers and in the Secondary CO Converter and must be removed.

Even a few parts per million of sulfur will poison the reforming catalyst and reduce the activity.

Therefore, every effort should be made to keep the sulfur concentration exit the desulfurization section as low as possible.

The natural gas, containing hydrogen required for hydrogenation which is recycled from the Syn Gas Compressor (K-301), is heated up to 400°C at the Fired Gas Heater (H-101) and fed to the Sulfur Removal Vessel (R-102AB). The inlet temperature of the Sulfur Removal Vessel is controlled by TIC-102.

Passing through the hydrogenation and sulfur absorption catalyst in the Sulfur Removal Vessel (R-102AB) the content of sulfur in the gas will be reduced to below 0.2 ppm by weight.

From the desulfurization unit the Natural Gas flows to the Gas Reform Section.

b. Primary Reforming

The desulfurized natural gas leaving the Sulfur Removal Vessel flows at a temperature of about 385°C through the flow control valve FCV-105. The gas is mixed with superheated steam of 380°C. The steam flow is regulated by the flow controller FRCA-103.

Both flow controllers are provided with low flow alarms. For start-up and partial load operation, the low range indicators FI-HC-106 and FI-HC-104 are provided.

The steam/gas mixture at a pressure of 36 Kg/cm^{2}G flows to the Steam Natural Gas Preheater (E-101), where the mixture is heated by the flue gas from the Primary Reformer to 520°C, recorded by TRA-121-3.

From the Steam Natural Gas Preheater the steam/gas mixture passes through the inlet manifold to the catalyst tubes of the Primary Reformer.

The Primary Reformer is the Topsoe type furnace which is an induced draft box heater having 200 catalyst filled tubes vertically installed in two separate chambers.

At the two combustion chambers fuel natural gas is fired by the radiant type wall burners, which are mounted equidistantly in six horizontal rows in the two vertical and longitudinal walls of each chamber.

This arrangement and the large number of wall burners provide an easy control of the desired temperature profile along the height of the reformer tubes and a uniform heat distribution throughout the length of the combustion chambers. The wall burners are fed with natural gas and they are self-aspiring ambient air for the combustion. Part of the fired heat released from the burners is transferred to the reformer tubes by radiation and by convection from the flue gas, while the remaining part of the fired heat (except heat losses from the combustion chambers to the surroundings) will leave the combustion chambers with the flue gas to the waste heat recovery section.

The flow of fuel gas to the burners is regulated by PIC-106, PIC-115, PIC-117. The two lower rows of burners are fed by a separate fuel gas line in order to maintain a partial firing of the Primary Reformer, when the Reformer is tripped.

At normal operation, the fuel gas pressure at the burners will be about 1.1 Kg/cm^2G. ~6 kg

The temperature rise of the gas inside the tubes will be slow only as most of the heat will be absorbed by the reaction between hydrocarbon and steam producing hydrogen, carbon monoxide, and carbon dioxide. The process gas leaving the tubes will be at a pressure of 31 Kg/cm^2G and a temperature of 795°C, and the residual methane content on dry basis will be 9.86% by volume, while other hydrocarbons are practically nil.

The outlet temperatures are recorded by TR-101, 1 to 5 one temperature measurement for each outlet collector. The differential pressure across the reformer tubes is indicated on PdR-104, and the inlet pressure is recorded by PR-103.

From the outlet collectors - five collectors are provided for each chamber - the process gas flows through the bricklined transfer line to the Secondary Reformer. The gas temperature in the transfer line is recorded by TRA-101-6, and furthermore a high temperature alarm is installed.

The flue gas leaves the two combustion chambers of the Primary Reformer at a temperature of about 1030°C. The temperature is recorded by TRA-101-8 provided with a high temperature alarm. The sensible heat of flue gas is utilized for several purposes in the waste heat recovery section of the Reformer.

The waste heat recovery section contains the Steam Natural Gas Preheater (E-101), Process Air Preheater (E-102), two Turbine Steam Superheaters (E-103AB), Flue Gas W.H. Boiler (E104) and two BFW Economizers (E-105AB).

The superheated steam temperature (TICA-156) controlled by the auxiliary burners provided at the upstream of the Turbine Steam Superheater. The flue gas is at last exhausted by the Induced Draft Fans (K-101AB) through the stack to the atmosphere.

Induced Draft Fans establish the draft slightly below the atmospheric pressure inside the combustion chambers of the Primary Reformer.

This pressure ensures against leakage of flue gas to the surroundings and is controlled by PIC-102. The content of oxygen in the flue gas is recorded by AR-102.

One of the Induced Draft Fans is driven by a back pressure steam turbine and the other by an electric motor.

c. Secondary Reforming

Effluent from the Primary Reformer at 795°C passes into the Secondary Reformer (R-103). While the air coming from the Process Air Compressor (K-302) passes the Process Air Pre-heater (E-102) located in the waste heat recovery section of the Primary Reformer. The air is preheated to 550°C, which temperature is recorded by TRA-121-4.

The preheated air enters the top of the Secondary Reformer through a specially designed burner which ensures a thorough mixing with the process gas.

The process gas and the air react in a partial combustion, thereby raising the temperature to approximately 1100 - 1150°C.

The exact air flow rate will principally be determined from the required 3 to 1 hydrogen-nitrogen ratio in the ammonia synthesis feed gas, so controlled by FRCA-107.

The resulting gas mixture flows downwards through the catalyst bed containing reforming catalyst, where the reforming takes place, resulting in a simultaneous decrease in temperature and methane content (methane reacts with steam to H_2, CO and CO_2). After passage of the catalyst bed, the temperature and the pressure are 969°C and 30 Kg/cm^2G respectively, whereas the methane content is reduced to 0.3% by volume on dry basis.

2.3 PURIFICATION

Purification involves the conversion of CO to H_2 and CO_2, the removal of the CO_2 and the conversion of the residual carbon oxides to methane by the methanation process.

a. Primary CO Conversion

The effluent stream from the Secondary Reformer at 969°C enters shell side of the Reform Gas Waste Heat Boiler (E-108) where it is

cooled to 365°C while producing 71 Kg/cm^{2}G steam. Then the Process Gas enters the Primary CO Converter (R-104) where the gas stream passes over a conventional iron and chromium catalyst bed where about 75% of the carbon monoxide (CO) reacts exothermically with the steam contained in the gas stream to form more hydrogen and carbon dioxide.

In the Primary CO Converter the carbon monoxide content of the gas stream is reduced to approximately 2.8%. The CO content of the Primary CO Converter effluent is controlled by adjusting the temperature of the inlet gas. The lowest temperature compatible with design CO leakage should always be used.

Primary CO Converter effluent is split with the major portion of the flow passing through the Converted Gas Waste Heat Boiler (E-109) utilizing the process heat to produce approximately 71 Kg/cm^{2}G steam. The smaller portion passes through the tube side of the Methanation Preheater (E-107) and heats the process feed gas to Methanator (R-106). The proportion of flow to each Exchanger is determined by the Methanator inlet temperature. The two flows unite again and pass through L.P. Boiler (E-110) which utilizes the process heat to produce 12 Kg/cm^{2}G steam while controlling the temperature of the stream to the Secondary CO Converter.

b. Secondary CO Conversion

The cooled gas by the L.P. Boiler at 215°C enters the Secondary CO Converter (R-105) and passes through a bed of sulfur guard to remove all traces of sulfur before it passes through the catalyst bed in which the carbon monoxide concentration is reduced to 0.33% by conversion of the carbon monoxide to carbon dioxide.

The effluent from the Secondary CO Converter flows to the CO_2 Removal Section before passing to the Methanator (R-106) for final treatment in the Gas Reform Section.

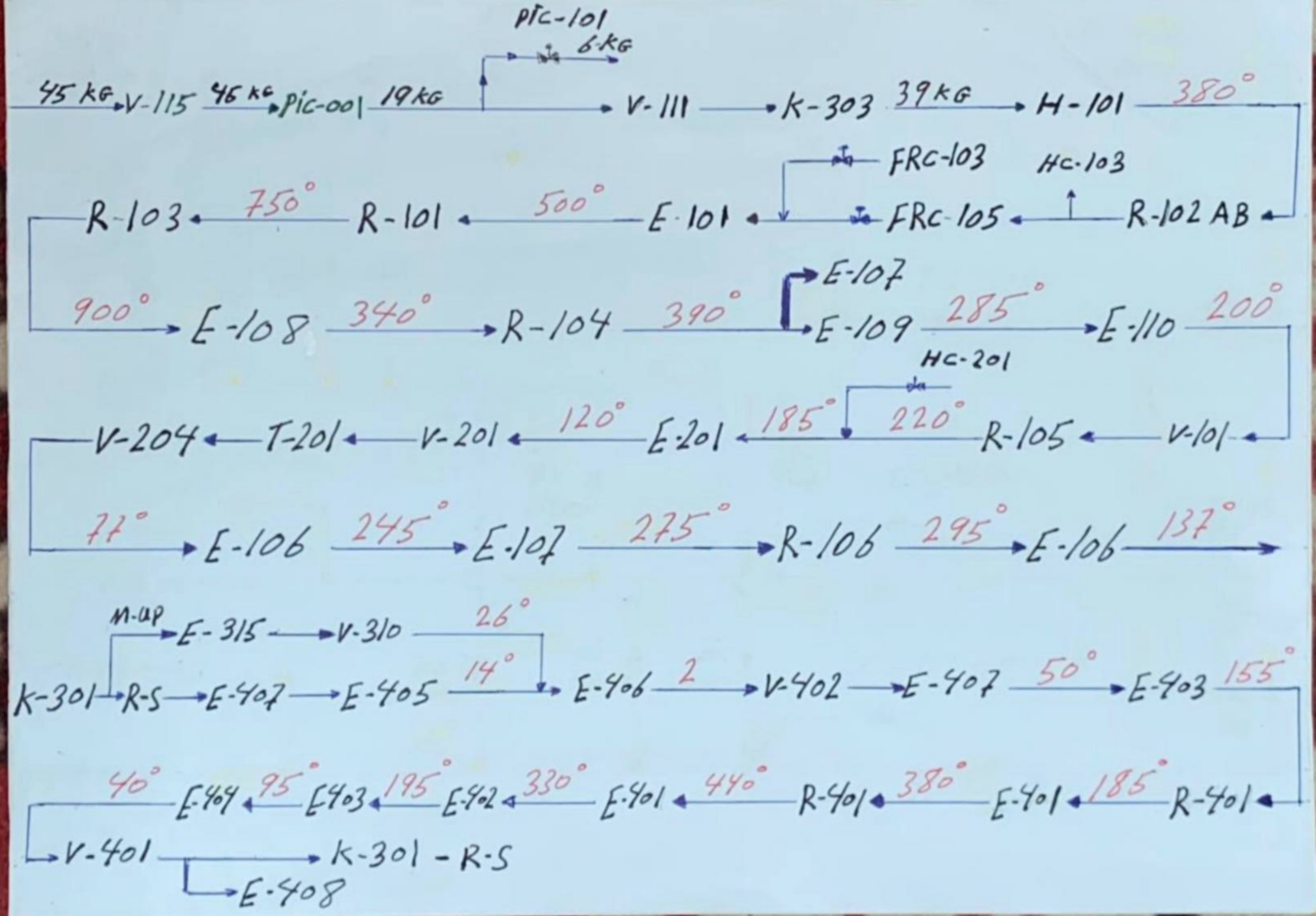
The Secondary CO Converter is provided with the inert gas circulation loop and the synthesis gas injection line which are used for the reduction of the catalyst at start-up.

A bypass connection around the Secondary CO Converter serves the two purposes;

- Bypassing the Secondary CO Converter during the reduction of the Secondary CO Converter catalyst at start-up.

- Bypassing the Converter to permit reduced capacity operation of the plant in case the Converter has to be taken out of service because of catalyst poisoning.

Due to the sensibility of the catalyst to high temperature the Converter is equipped with a series of temperature indicators to determine the temperature throughout the catalyst bed.


PIC-101
6 kG
45 kG
V-115
46 kG
PIC-001
19 kG
V-111
K-303
39 kG
H-101
380°
FRC-103
HC-103
R-103
750°
R-101
500°
E-101
FRC-105
R-102 AB
E-107
900°
E-108
340°
R-104
390°
E-109
285°
E-110
200°
HC-201
V-204
T-201
V-201
120°
E-201
185°
220°
R-105
V-101
77°
E-106
245°
E-107
275°
R-106
295°
E-106
137°
M-up
E-315
V-310
26°
K-301
R-5
E-407
E-405
14°
E-406
2
V-402
E-407
50°
E-403
155°
40°
E-404
95°
E403
195°
E-402
330°
E-401
44°
R-401
380°
E-401
185°
R-401
V-401
K-301 - R-5
E-408